CMP4221 Companion Study Deck

DWT Encryption backup explanations

Use this for rehearsal, defense, and Q&A. Do not present all of it in the 10-minute slot.

Requirement proof

Requirement	Evidence
Journal	ACM Transactions on Multimedia Computing, Communications, and Applications
Year	Crossref: online 2026-02-27; print 2026-03-31
DOI	10.1145/3769123
Main deck timing	10 slides; Abdul 1-5; Maria 6-10
Required content	Novelty slide 4; methodology slides 5-7; findings slides 8-9

Paper identity

- Title: Multi-Image Encryption Scheme Based on Chaotic Pseudo-Random Signal Generator and DWT Compression.
- Authors: Yidan Xu, Suo Gao, Yinghong Cao, Jun Mou.
- Article 82, pages 1-20, ACM TOMM Vol. 22 Issue 3.
- Core topic: compress multiple color images, merge them into a cube, then encrypt with chaos.
- Do not use Options/3762860.fm.pdf as the article source; it is issue front matter.

Problem framing

Bandwidth

Multi-color images are large because every pixel has R, G, and B values.

Security

Visible image content must become unreadable before transmission.

Batch use

Multiple images need one efficient workflow instead of repeated single-image encryption.

DWT formula and LL-only compression

- For each channel: [LL, LH, HL, HH] = DWT(channel).
- LL = low-frequency approximation. It keeps most visible structure.
- LH, HL, HH = detail bands. They carry edge/detail information.
- Keeping LL only gives half height and half width, so area becomes 1/4.
- Recovery is lossy, but the paper reports PSNR above 32 dB.

Chaotic key parameters

- The chaotic map is deterministic: same key gives the same sequence.
- It is sensitive: a tiny initial-value change gives a totally different sequence.
- Paper parameters include a, b, c, d, k_1, k_2 and initial values x_0, y_0, z_0.
- These sequences drive both position swaps and XOR diffusion.
- Security claim depends on randomness tests and key sensitivity.

Full pipeline recipe

- 1. Split every color image into R, G, B channels.
- 2. Apply DWT to each channel and keep LL.
- 3. Stack all LL data into cube C.
- 4. Generate plaintext-related chaotic parameters.
- 5. Confusion swaps positions; diffusion changes values.
- 6. Transmit cipher cube D; receiver reverses the steps.

Cube sizing and zero-padding

- Each compressed image has size hi/2 by wi/2 by li.
- The cube width and height follow the largest compressed images in the batch.
- Smaller images need padding so all data fits into the shared cube shape.
- Limitation: padding takes extra space and reduces efficiency for mixed-size batches.
- Future-work line: reduce extra-space utilization for more robust, less resource-intensive encryption.

Confusion algorithm

- Purpose: change where pixels are.
- Input: plaintext cube C.
- Chaotic sequences become coordinate offsets S1, T1, U1.
- For each pixel (i, j, k), compare coordinates against offsets.
- Nine swap cases move pixels into new cube positions.
- Output: C-prime, a position-scrambled cube.

Diffusion algorithm

- Purpose: change pixel values.
- Flatten C-prime into vector V.
- First value XORs with a chaotic seed.
- Later values use index mod 3 to select X2, Y2, or Z2.
- Each output also depends on the previous output, creating a cascade.
- Output: cipher cube D.

Metric: PSNR

- PSNR = Peak Signal-to-Noise Ratio.
- Unit: decibels.
- Higher PSNR means reconstructed image is closer to the original.
- The paper says around 30 dB is generally acceptable visual recovery.
- Reported result: 32.0601 dB after compression and encryption/decryption.

PSNR comparison

Scheme	PSNR (dB)
Proposed scheme	32.0601
Ref. [8]	26.2896
Ref. [22]	27.6917
Ref. [46]	27.3366
Ref. [19]	27.4886

NIST randomness summary

- NIST suite checks whether a pseudo-random generator behaves statistically random.
- The paper reports all 15 tests passed.
- Pass condition used: p-value >= 0.01 and pass rate above the accepted threshold.
- Presentation line: the chaotic generator passed the randomness battery.

Differential attack metrics

Metric	Target	Paper result	Meaning
NPCR	99.6094%	99.6533%	How many pixels change after a tiny plaintext change
UACI	33.4635%	33.4887%	Average intensity change between two cipher images

Information entropy

- Information entropy measures how uniform/unpredictable the cipher pixel distribution is.
- The theoretical maximum for 8-bit images is 8.
- The paper reports 7.9994, essentially maximum entropy.
- Presentation line: the cipher image is statistically close to uniform noise.

Speed and efficiency

Step	Time (s)	Speed
Confusion	0.1124	177.9359
Diffusion	3.1381	6.3733
Total encryption	3.2505	6.1529

Likely Q&A

- Q: Is this lossless? A: No. DWT LL-only compression is lossy, but PSNR is still strong.
- Q: Why use a cube? A: It lets the method encrypt multiple images together in one object.
- Q: What is the main limitation? A: Zero-padding wastes space for mixed-size images.
- Q: Is chaos alone enough? A: No. The scheme combines confusion, diffusion, and tested randomness.
- Q: What is the strongest result? A: PSNR 32.0601 dB plus strong NPCR/UACI/entropy values.

Rehearsal checklist

- Abdul: keep slides 3 and 5 tight; do not over-explain wavelets.
- Maria: define confusion and diffusion in one sentence each before details.
- Both: say numbers cleanly: PSNR 32.06, NPCR 99.65, UACI 33.49, entropy 7.9994.
- Do one timed run under 10 minutes before presenting.
- Use the study deck only for backup and Q&A practice.